

Zhuoyuan Hao

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Experience with large-scale model training at Tencent, research experience with 2 (co-)first-author papers (1 ICLR poster, 1 under review)

Education

Harbin Institute of Technology (Shenzhen)

Shenzhen, China

COMPUTER AND ELECTRONIC COMMUNICATIONS CLUSTER - COMPUTER SCIENCE AND TECHNOLOGY

Sep. 2023 – Dec. 2026

- Selected for the **Integrated Graduation Project** program — completing the full undergraduate thesis in **half a year** (two semesters ahead of schedule)
- English proficiency **IELTS 7.0**; Weighted GPA 83

Internship Experience

Tencent IEG

Technology Operations

RESEARCH INTERN

Jun. 2025 – Sep. 2025

- Achievement: Model integrated into the internal operations environment of the business group, with community version **open-sourced to modelscope** by China Academy of Information and Communications Technology (StabilityLab/zhiyu)
- Personal contributions:
 - Post-training: Built **fault injection** and **tool invocation framework** in operations environment; explored Reward design; **SFT-COT distillation** to restore reasoning capabilities
 - Continual Pre-Training data exploration: Data quality assessment, data rephrasing, **synthetic data selection & evaluation**

Tencent Codebuddy Agent-SFT

Shenzhen

RESEARCH INTERN

Feb. 2026 – Mar. 2026

- Data mixture for Agentic Coding user SFT: designed mixing strategies and evaluation **rubrics** during data refinement; knowledge-distilled **Opus-4.5** into an 8B model
- How to train rollout traces: traces come with Claude-rewritten thinking; keep train-serve consistency by applying loss only to **model-generated tokens**:
 - /think vs thinking mode: the /think terminator is decided by the model, while <think> is baked into the template. Non-thinking and thinking templates differ, so non-thinking data is not trained. Under enable-thinking, /think still takes loss even if reasoning is empty — the rewrite is a small safety-filter model; if Opus only thought a few tokens and the rewrite skipped them, that span is treated as unimportant, and an empty /think is more efficient
 - ms-swift: the default trains <think> template tokens as well, breaking train-serve consistency; custom loss scale sets their weight to 0
 - Loss mask: Agent SFT defaults to raw trace → chat-template token sequence → per-token loss mask. The stable choice so far is loss on model-generated tokens; slime and llamafactory differ in the details and there is no consensus yet

Research Experience

Echo of Prompt: Understanding CoT Reasoning in LLMs

Prof. Jing Li, HITSZ

FIRST AUTHOR – ICLR 2026

May 2025 – Nov. 2025

- Motivation: LLM CoT traces consistently begin by restating the question (**Echo**). Echo often co-occurs with the emergence of reasoning ability — e.g., CoT capability elicited via prompting, and the “aha moment” of DeepSeek-R1-zero trained with RLVR
- Mechanistic Analysis: (1) Quantify the probabilistic cost-benefit relationship of Echo; (2) Reveal Echo’s role in **Attention Refocusing**; (3) Establish causal evidence between Echo and accuracy; (4) Case study: information flow visualization and **identification of additionally attended tokens**
- Method Transfer: (1) **SFT** — construct high-quality CoT datasets, derive an Echo-free version via MLP classifiers and model verification, finding **severe performance degradation** without Echo; (2) **Prompt-based method** — insert a “review the question” cue at inference time, achieving significant improvement over baseline under controlled token budgets

Reward Hacking in Rubric-based RL with LLM Judge

CO-FIRST AUTHOR

Prof. Xiaozhi Wang, Tsinghua

Nov. 2025 – May 2026

- Background: In open-ended tasks, LLM-as-judge is used as the RL reward signal, but policy models may exploit latent biases in the LLM judge to obtain high rewards that do not align with human expectations
- Challenge: In practice, such reward hacking is covert and intertwined with multiple LLM biases, making analysis, detection, and mitigation extremely difficult
- Controllable Hacking Environment: Inject known biases into the LLM judge to stably reproduce reward hacking, observe reward divergence, and precisely locate hacking onset; characterize hacking along two axes — **discoverability** and **exploitability**
- Detection Agent: **Automatically detect** reward hacking onset from training logs, characterize hacking behavior patterns, and provide **interpretable diagnostic evidence**

Project Experience

Date-Match

CO-FOUNDER & ALGORITHM LEAD

MiraclePlus Referral

Feb. 2026 – Present

- Led end-to-end product and algorithm development from inception; designed the core questionnaire-based compatibility matching system powering a social relationship app (date-match.online) targeting young users
- Achieved strong early traction: collected **100,000+ valid questionnaires within 10 days** of launch and reached a planned user base of **170,000+ within the first month**, demonstrating clear product-market fit
- Drove **organic viral growth across university campuses** nationwide — student communities at Fudan, Tsinghua, Wuhan University, Tianjin University, and others independently replicated the matching model, generating campus-wide social propagation
- Pioneered the “questionnaire-driven intelligent matching” product category in China; received an internal referral from **Miracle-Plus** (Y Combinator’s China counterpart)
- Tech lead for a 7-person engineering team (frontend, backend, algorithms) in a 14-person org; introduced AGENTS.md as the AI development entry, issue-to-PR workflow, Copilot code review, and Codex-based architecture review

Extracurricular Activities

University Rowing Team - Men’s Eight

MAIN ROWER - POSITION 7 (LEAD STROKE)

HITSZ

Sep. 2023 – Jun. 2024

- **Won championships** at Shenzhen city and local municipal level competitions
- Developed teamwork, leadership, and time management skills

University International Exchange Association

MEMBER

HITSZ

Sep. 2023 – Present

- Participated in organizing multiple international cultural exchange activities
- Contributed to writing the “**Study Abroad Handbook**” and building its database platform, providing services to fellow students